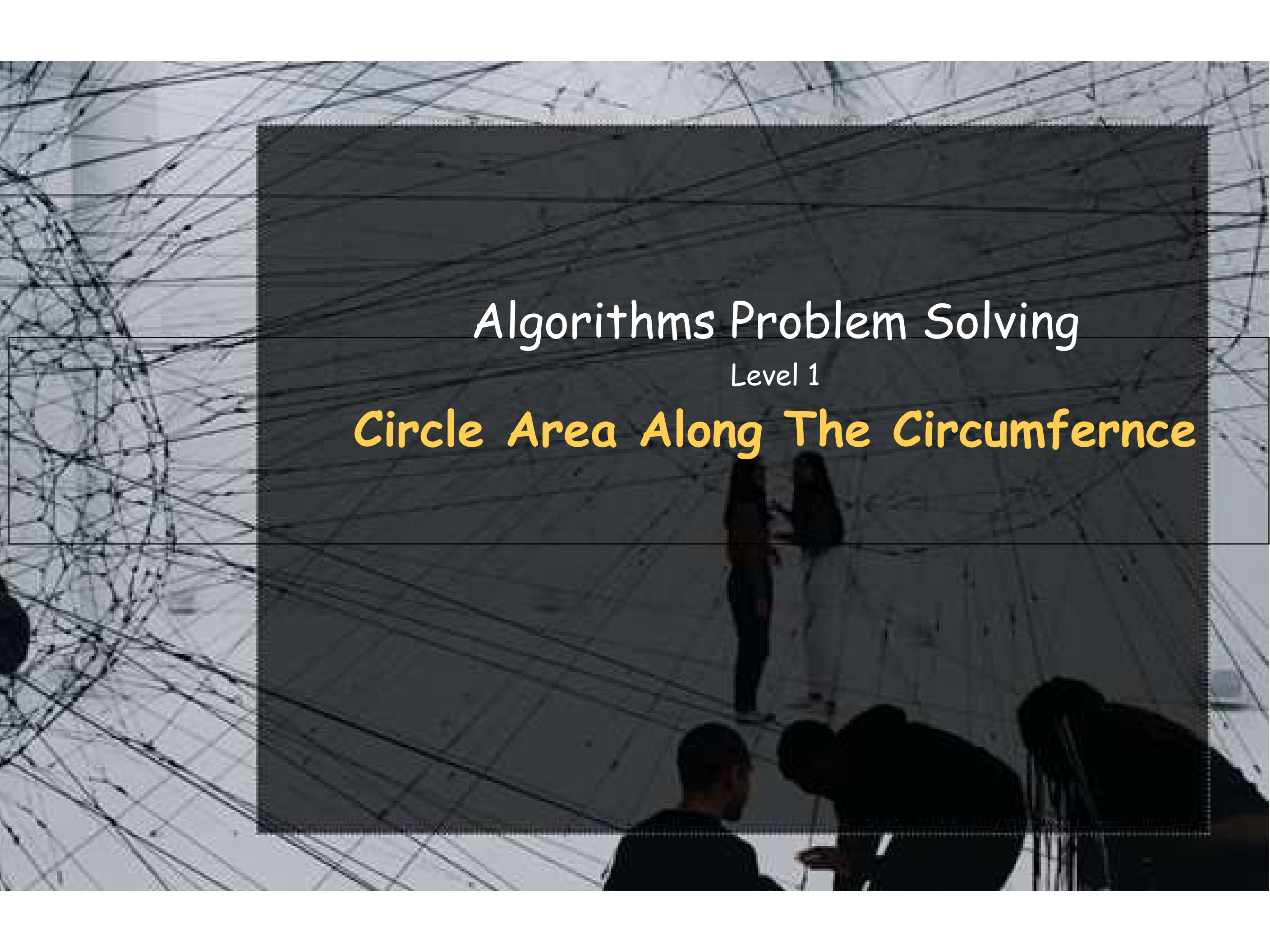


# Algorithms Problem Solving Level 1

Circle Area Along The  
Circumference



# Algorithms Problem Solving

Level 1

## Circle Area Along The Circumference

## Definition

- ◆ This program calculates the area of a circle when its circumference is given.
- ◆ The program uses the circumference formula,  $\pi$ , `pow()`, division, functions, and float values.
- ◆ The area can be calculated directly from the circumference without first calculating the radius.

## Explanation

- ◆ The `readCircumference()` function reads the circumference  $L$  from the user and returns it as a float.
- ◆ The circumference formula is:
  - ◆  $C = 2 \times \pi \times R$
  - ◆ Therefore:
    - ◆  $R = C / (2 \times \pi)$
  - ◆ The circle area formula is:
    - ◆  $\text{Area} = \pi \times R^2$
  - ◆ By substituting the radius formula, we get:
    - ◆  $\text{Area} = C^2 / (4 \times \pi)$
  - ◆ `CircleAreaByCircumference()` uses this formula directly.
  - ◆ `pow(L, 2)` calculates the circumference squared.
  - ◆ `M_PI` represents  $\pi$ .
  - ◆ `printResult()` receives the calculated area and prints it.
  - ◆ In `main()`, the functions are nested:
    - ◆ `printResult(CircleAreaByCircumference(readCircumference()))`;
    - ◆ `readCircumference()` runs first, its result is passed to `CircleAreaByCircumference()`, and the calculated area is passed to `printResult()`.

## Steps in main()

1. Read the circle circumference  $L$ .
2. Calculate  $L^2$  using `pow()`.
3. Calculate  $4 \times \pi$ .
4. Divide  $L^2$  by  $4 \times \pi$ .
5. Return the calculated area.
6. Pass the returned area to `printResult()`.
7. Print the circle area.

# 💡 Example

◆ If the user enters:

Circumference: 20

◆ Calculate:

$$\text{Area} = 20^2 / (4 \times \pi)$$

$$\text{Area} = 400 / (4 \times \pi)$$

◆ The result is approximately 31.831.

➡ Result

Circle Area = 31.831

## Note

- ◆ The formula used in the program is:
- ◆  $\text{Area} = C^2 / (4 \times \pi)$
- ◆ This formula is derived from the relationship between circumference, radius, and circle area.

## ⊘ Warning

- ◆ `readCircumference()` returns a float, but `CircleAreaByCircumference()` receives an int parameter.
- ◆ If the user enters a decimal circumference such as 20.5, the decimal part will be lost when the value is passed to `CircleAreaByCircumference()`.
- ◆ A better parameter type would be:
- ◆ `float CircleAreaByCircumference(float L)`



# Comparison

## ▨ Using Radius

- ✓ Formula:  $\text{Area} = \pi \times R^2$ .
- ✓ The radius is given directly.
- ✓ The calculation uses the radius squared.

## ▤ Using Circumference

- ✓ Formula:  $\text{Area} = C^2 / (4 \times \pi)$ .
- ✓ The circumference is given directly.
- ✓ The radius does not need to be calculated separately.

## ✦ Summary

- ✓ A circle's area can be calculated from its circumference.
- ✓ The formula is  $C^2 / (4 \times \pi)$ .
- ✓ `pow(L, 2)` calculates the circumference squared.
- ✓ The formula is derived from  $C = 2 \times \pi \times R$  and  $A = \pi \times R^2$ .
- ✓ Matching float parameter types prevents decimal values from being lost.

Thank you 😊